

YTU Department of Mathematical Engineering
MTM2522 Data Management and File Structures 1

Homework

Due: May 31, 2026 23:59

Title:

Design and Implementation of a Scalable Multi-User Database System for Smart Healthcare Management

 Project Objective

Design and implement a **complete database system** for a **Smart Healthcare Platform** that supports:

- Multi-user access (patients, doctors, admins)
- Complex data relationships (EER level)
- Query optimization and indexing
- Real-world constraints and normalization

This project directly satisfies course outcomes like:

- ER/EER modeling
 - Relational schema design
 - Functional dependencies & normalization
 - SQL querying
 - File structures & indexing
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 Project Scenario

You are developing a database system for a **large hospital network** that includes:

- Multiple hospitals and departments
- Doctors with different specialties
- Patients with medical histories

- Appointments, prescriptions, and lab results
- Insurance and billing system

The system must support **efficient querying, integrity, and scalability**.

Project Tasks (Step-by-Step)

1. Requirements Analysis

Define:

- Entities (Patient, Doctor, Hospital, Appointment, etc.)
- Relationships (Doctor–Patient, Patient–Appointment, etc.)
- Constraints (e.g., one patient can have many appointments)

 Deliverable:

- Written requirement document (2–3 pages)
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2. ER Model Design

Create a detailed **ER Diagram** including:

- Entities & attributes
- Primary keys
- Relationships (1-1, 1-N, N-M)
- Participation constraints

 Must include:

- Weak entities (e.g., dependent records like prescriptions)
 - Composite attributes
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3. Enhanced ER (EER) Model

Extend your ER model using:

- Specialization / Generalization
 - Example:
 - User → Patient / Doctor / Admin

- Inheritance
- Categories (union types if needed)

👉 Deliverable:

- EER Diagram + explanation
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◆ 4. Mapping to Relational Model

Convert ER/EER into tables:

- Define primary keys & foreign keys
- Handle:
 - Multivalued attributes
 - Weak entities
 - N:M relationships

👉 Deliverable:

- Relational schema (table structures)
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◆ 5. Functional Dependencies

Identify FDs for at least 3 major tables.

Example:

PatientID → Name, BirthDate, Phone

DoctorID → Name, Specialty

👉 Deliverable:

- FD list + explanation
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◆ 6. Normalization

Normalize your schema up to:

- 1NF
- 2NF
- 3NF

- BCNF (at least for critical tables)

👉 Show:

- Before & after tables
 - Decomposition steps
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◆ 7. SQL Implementation

Use **MySQL / PostgreSQL / SQL Server**

Must include:

- CREATE TABLE (with constraints)
 - INSERT data (at least 20–30 records)
 - ALTER operations
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◆ 8. Advanced SQL Queries

Write **at least 10 complex queries**, including:

- JOIN (INNER, LEFT)
- GROUP BY + HAVING
- Subqueries
- Nested queries

Example:

- Find doctors with more than 10 patients
 - List patients with no appointments
 - Get most common diagnosis
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◆ 9. Relational Algebra & Calculus

Write **5 queries** in:

- Relational Algebra
 - Tuple Relational Calculus
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◆ 10. Indexing & File Structures

Explain and implement:

- Indexes:
 - B+ Tree index
 - Hash index
- Compare:
 - Indexed vs non-indexed query performance

👉 Deliverable:

- Performance discussion
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Expected Deliverables

1.  Report (15–25 pages)
 2.  ER + EER diagrams
 3.  SQL scripts
 4.  Query outputs
 5.  Normalization steps
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Evaluation Rubric (Suggested)

Component	Weight
ER/EER Design	20%
Relational Mapping	15%
Normalization	15%
SQL Implementation	20%
Queries	15%
Indexing & Analysis	10%